Robotic Manipulation

Perception, Planning, and Control

Russ Tedrake

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Last modified 2026-10-11.
How to cite these notes, use annotations, and give feedback.

Note: These are working notes used for a course being taught at MIT. They will be updated throughout the Fall 2026 semester.

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Appendix B
Drake

Drake is the main software toolbox that we use for this text, which in fact originated in a large part due to the MIT Underactuated Robotics course. The Drake website is the main source for information and documentation. The goal of this chapter is to provide any additional information that can help you get up and running with the examples and exercises provided in these notes.

B.1 Pydrake

Drake is primarily a C++ library, with rigorous coding standards and a maturity level intended to support even professional applications in industry. In order to provide a gentler introduction, and to facilitate rapid prototyping, I've written these notes exclusively in python, using Drake's python bindings (pydrake). These bindings are less mature than the C++ backend; your feedback (and even contributions) are very welcome. It is still improving rapidly.

In particular, while the C++ API documentation is excellent, the autogenerated python docs are still a work in progress. I currently recommend using the C++ documentation to find what you need, then checking the Python documentation only if you need to understand how the class or method is spelled in pydrake.

There are also a number of tutorials in Drake that can help you get started.

B.2 Online Jupyter Notebooks

I will provide nearly all examples and exercises in the form of a Jupyter Notebook so that we can leverage the fantastic and relatively recent availability of (free) cloud resources.

B.2.1 Running on Google Colab

Follow an "Open in Colab" link beside an example and run the cells in order. The first code cell installs the required packages. Save a copy in Google Drive if you want to keep your edits.

Many notebooks use Meshcat for interactive visualizations. After running `StartMeshcat()`, click "Open Meshcat in a separate window" and keep the notebook open while using the viewer. To embed the viewer beside your current cell, run:

```
from manipulation.meshcat_utils import DisplayMeshcat
DisplayMeshcat(meshcat)
```

B.2.2 Enabling licensed solvers

You can enable more powerful solvers for `MathematicalProgram` if you have a license (most are free for academics). Please see the MOSEK and Gurobi documentation for instructions.

B.3 Running on your own machine

As you get more advanced, you will likely want to run (and extend) these examples on your own machine. On platforms/configurations that Drake supports (the latest two releases of Mac and Ubuntu using the system default python versions). The simplest installation is via `pip`; I would typically recommend using a virtual environment:

Copy code

```
python3 -m venv venv
source venv/bin/activate
python -m pip install "manipulation[all]" jupyterlab
```

Then download notebooks from the book directory on GitHub and run them with `jupyter lab` (or develop your own notebooks).

The Drake website also has a number of alternative installation options, including precompiled binaries and Docker instances. If you build from source (awesome!), then make sure you also build the python bindings. Finally, make sure you have the Drake installation in your `PYTHONPATH`.

If you want to download all of the notebooks at once, you can `git clone` the course notes repository. You'll likely want to start from the repository's root directory. Then launch your notebook with:

Copy code

```
jupyter notebook
```

The examples for each chapter that has them will be in a .ipynb file right alongside the chapter's html file, and the notebook exercises are all located in the `exercises` subdirectories.

B.4 Getting help

If you have trouble with Drake, please follow the advice here. If you have trouble with the manipulation repo, you can check for known issues (and potentially file a new one) here.

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